Department of Mathematics

ESI Building, Penryn

Harrison Building

Hope Hall

Geoffrey K. Vallis
Head of Department

Laver Building

Numbers

Academic Staff

66 FTE (approx)
16 (24%) female

77 total academic staff (E&R + E&S, including temporary appointments, and 20% appointments)

L (Lecturers): 19
SL: 21
AP: 13
P: 24

57 postdocs/researchers

REF 2021

UoA 10 (Maths):
48 staff submitted (**48 FTE**)

UoA 7 (Env. Sci):
25 staff submitted (**20 FTE**)
7 from joint Met Office positions.

30% of UoA-7 papers submitted from Maths.
(but only 23% of UoA-7 staff from Maths.)

Students

Total: 1033 (approx)

UG: 800

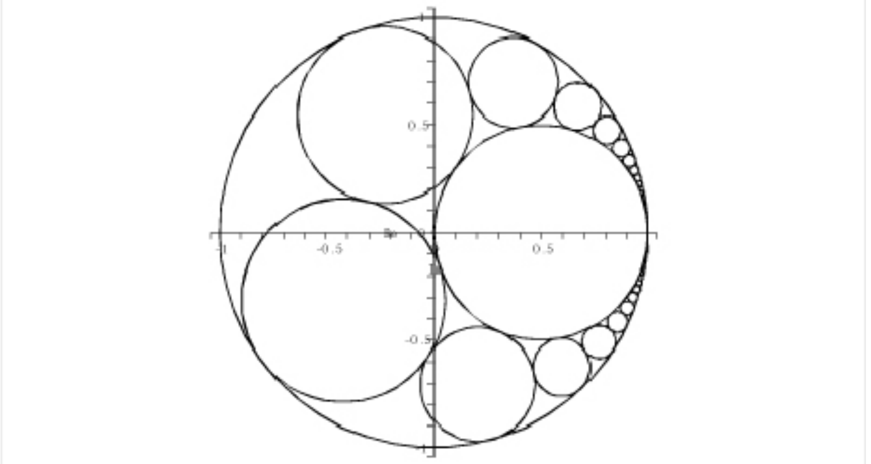
PGT: 107 (+ 30)

PGR: 126

Gender Split:

Female: 34.5%
Male: 65.2%
Other: 0.3%

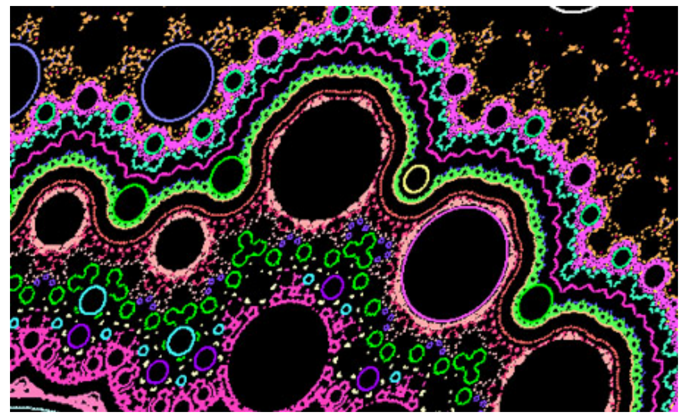
Number theory, Algebra, Geometry
8 Academic Staff



To advance mathematics itself.
Seek mathematical truth, prove theorems, etc
Applications to cryptography, combinatorics, finance.

Heilbronn Institute (GCHQ partnership),
Kyoto, Brown, Cornell Universities

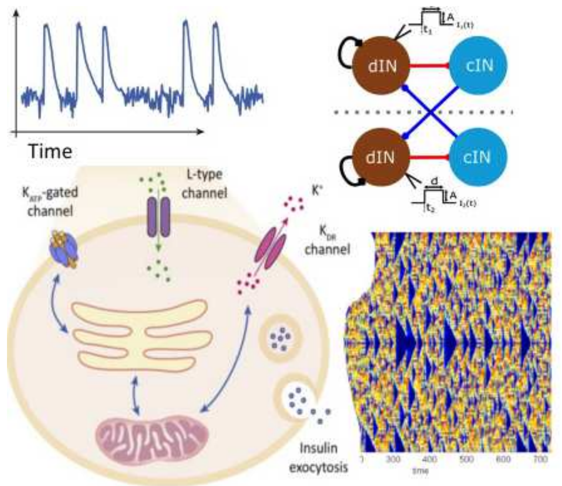
Dynamical Systems and Analysis
6 Academic Staff



ODEs, patterns, networks, maps, ergodic theory.
Pure leaning with applied spin-offs, namely Living Systems Mathematics and Environmental Mathematics.

Met Office, NHS,
Willis Reinsurance

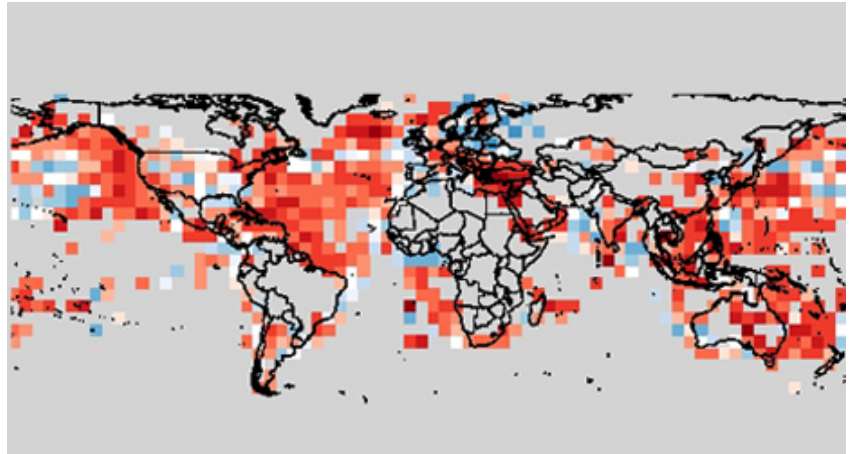
Living Systems Mathematics
9 Academic Staff



Applications to biology and medicine
Combines dynamical systems, data science, image processing and biology.

NHS, Medicines Discovery Catapault,
Braininhand, Brainbow (app developers),
Ludger, Certus (biomarker/clinical database development).

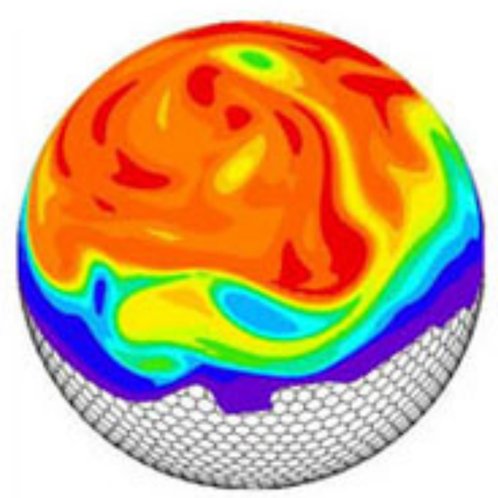
Statistics and Data Science
12 Academic Staff



Foundational Statistics and ethical AI,
Uncertainty quantification (UQ),
Applications to climate, weather, epidemiology

Turing Inst., Met Office, WHO, World Bank,
Willis Insurance, IPSOS Mori, CUHK,
KTP with Agile Applications, others.

Geophysical and Astrophysical Fluid Dynamics
13 Academic Staff



Core applied mathematics.
Theory of weather & climate, solar physics, MHD, planetary atmospheres.

Met Office (MOAP chair),
Harvard Smithsonian, Kyoto,

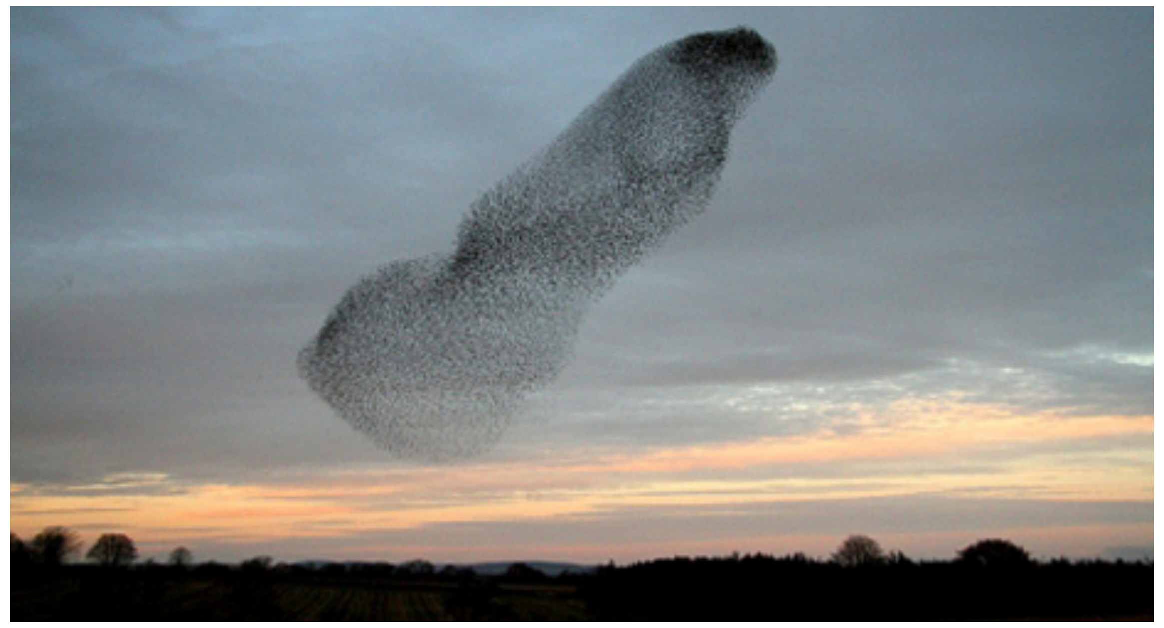
Climate Dynamics
17 Academic Staff
7 joint with Met Office

Processes in weather and climate:
aerosols, storms, carbon cycles, clouds, global warming, etc.

Met Office (7 joint positions, MOAP chair)
WMO and IPCC, WCRP.

Mathematics and the Environment, Penryn campus

6 Academic Staff



Applications of dynamical systems, control theory and complexity to ecology, epidemiology, human health, renewable energy, sustainability and business.

One of the most connected/collaborative groups within the University of Exeter:

InnovateUK: NJW Ltd (KTP), Hirst Magnetic Instruments Ltd (KTP), Chelonia Ltd (KTP);
Match-funded PhD studentships: NJW Ltd, Hirst Magnetic Instruments Ltd;
ERDF-funded collaborations: Agri-tech Cornwall, Cornwall New Energy, Deep Digital Cornwall, Marine-i, Smartline, Tevi (enabling collaborations with local/regional businesses, e.g., Bur-Bax-10 Ltd, Buzz Interactive Ltd, Coastline Housing Ltd, Numeration Systems Ltd, Newquay Hypnotherapy, Proper Cornish Ltd, The Smart Home Co. Ltd);
MSc projects with industrial partners: GridIMP Ltd, Hi9 Ltd, SWIN Ltd;
ERASMUS+ funded partnership: “Demain, l’ecole” (College Kercourat, Centre d'Animation Locale, College Saint Pol Roux (France), Scoala Gimnaziala «Aurel Decei» Gura Raului (Romania), Cornwall Neighbourhoods for Change Ltd.;
ERDF-funded project partners: Cornwall Council, Cornwall Development Company, Cornwall Wildlife Trust, Volunteer Cornwall, South West Academic Health Science Network;

New MSc Programmes in Applied Data Science, Modelling & Environment!

Institutes, Centres and Other Programmes

- Natural Sciences Programme (NatSci)
- Living Systems Institute (LSI)
- Global Systems Institute (GSI)
- Environmental Science Institute, Penryn (ESI)
- IDSAI (Data Science and Artificial Intelligence)
- Joint Centre for Excellence in Environmental Intelligence (with Met Office).
- Chinese University of Hong Kong (CUHK), joint centre for Environment, Sustainability and Resilience (ENSURE).
- Exeter Maths School.

Isaac Newton Institute for Mathematical Sciences

WCRP World Climate Research Programme

Willis Towers Watson

Heilbronn Institute for Mathematical Research

The Alan Turing Institute

Met Office

NPL

Newton Fund

académie Rennes
Région académique BRETAGNE

CATAPULT Medicines Discovery

Avon and Wiltshire NHS
Mental Health Partnership NHS Trust

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ICSHNet COST ACTION IS1408

NHS Devon Partnership NHS Trust

World Health Organization

EXETER MATHS SCHOOL
FREEDOM FOR THINKERS

IAS
TUM Institute for Advanced Study
Technische Universität München

FIOCRUZ

IOP Institute of Physics

CORNWALL COUNCIL

Cornwall Neighbourhoods for Change

Ministry of Justice

BULGARIAN ACADEMY of SCIENCES 1869

Collège ERZOURAT

CORNWALL Development Company

Coastline housing

Volunteer Cornwall

Recent fellowships, awards and indicators of success (partial list)

MATHEMATICS UNIVERSITY OF EXETER

Alan Turing Institute fellows
Prof. Peter Challenor
Prof. Mark Kelson
Prof. Gavin Shaddick
Prof. Krasimira Tsaneva-Atanasova
Prof. Danny Williamson

Royal Society Wolfson Awards
Prof. David Stephenson
Prof. Geoffrey Vallis
Prof. Peter Cox
Prof. Pierre Friedlingstein

Edward Appleton Medal of the Institute of Physics
Prof. Adam Scaife

Fellow, Bulgarian Academy of Sciences
Prof. Krasi Tsaneva-Atanasova

Lloyds Science of Risk Prize
Dr. Ben Youngman

Imagining the Future of Undergraduate STEM Education: US Nat'l Academies of Science, Education and Medicine
Dr. Layal Hakim

ERC Advanced Award
Prof. Peter Cox

ERC Early Career award:
Dr Kirsty Wan

MRC Skills Development Fellowships
Dr. Jen Creaser
Dr. Yolanda Hill

NERC Independent Research Fellow
Dr. George Efstathiou

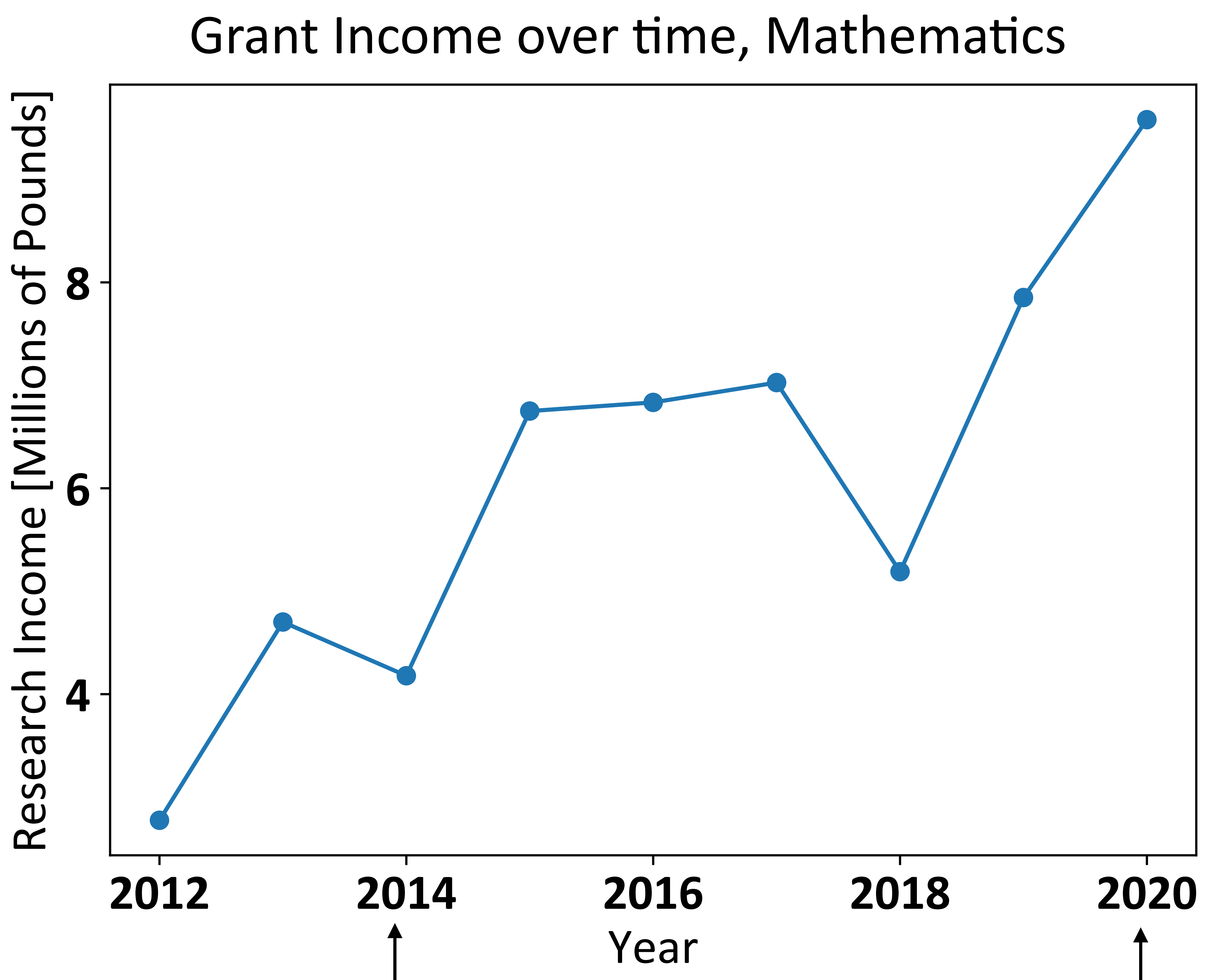
EPSRC Early Career Fellowship
Dr. Jan Sieber

Hans Fischer Fellowship (TU Munich-IAS)
Dr. Christian Bick, Prof. Krasi Tsaneva-Atanasova

STFC Ernest Rutherford Fellowship
(recently finished)
Prof. Andrew Hillier

Plus numerous plenary lectures at international meetings, international conferences hosted, invited distinguished lectures, etc.

Research Income Success



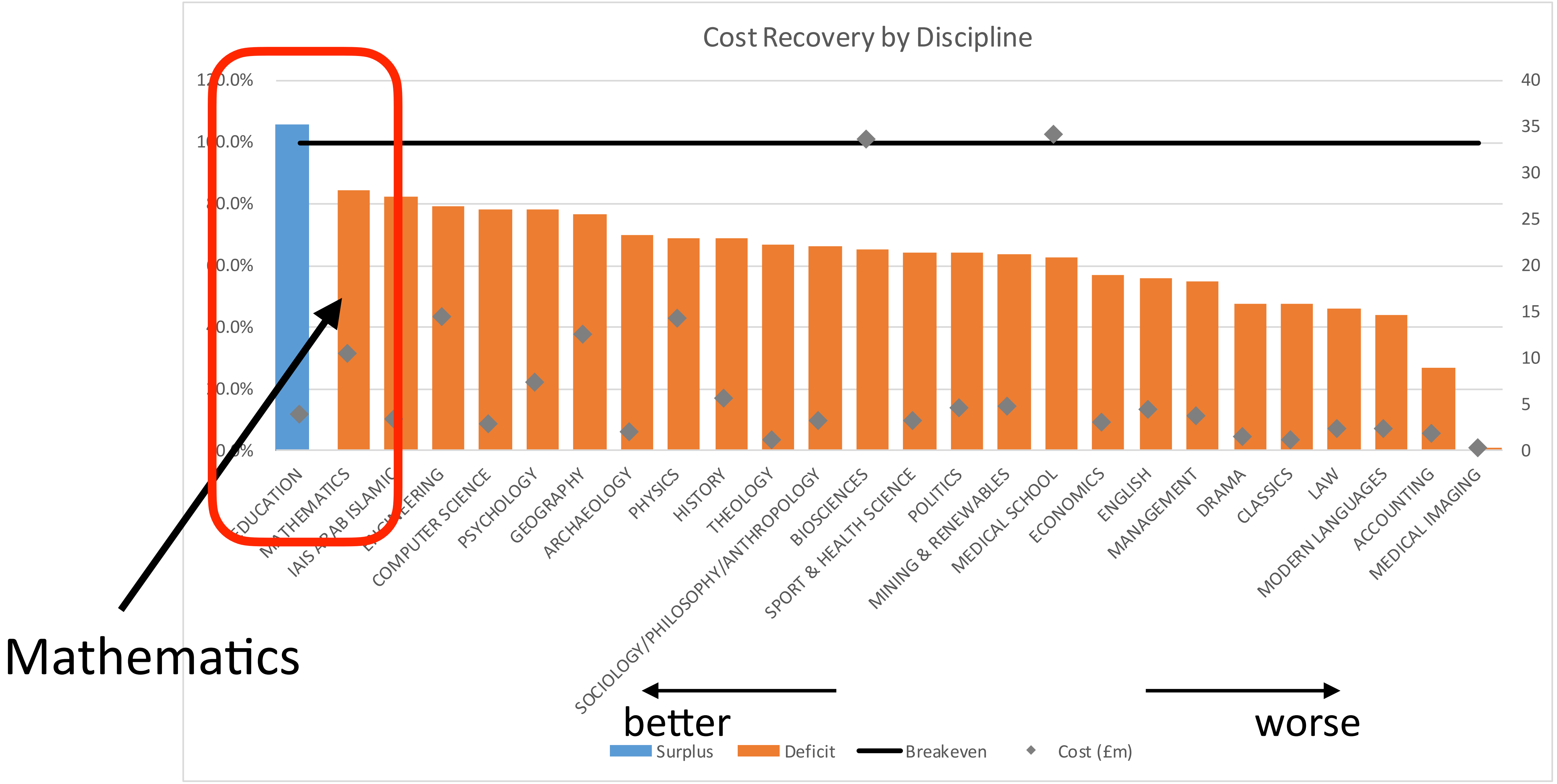
Academic Staff

34 UoA-10
9 UoA-7

48 UoA-10
20 UoA-7

1. Research income tripled over last 8 years
2. Income per FTE has increased.
3. UoA 10: **Income per FTE $\sim$ 1.7 x Russell group average**.

Cost recovery % by discipline (excl MSI) – research



Cost recovery and earnings (i.e. how much University makes) are high compared to other departments.

(NB: Teaching income in Mathematics also very high.)

Education Success

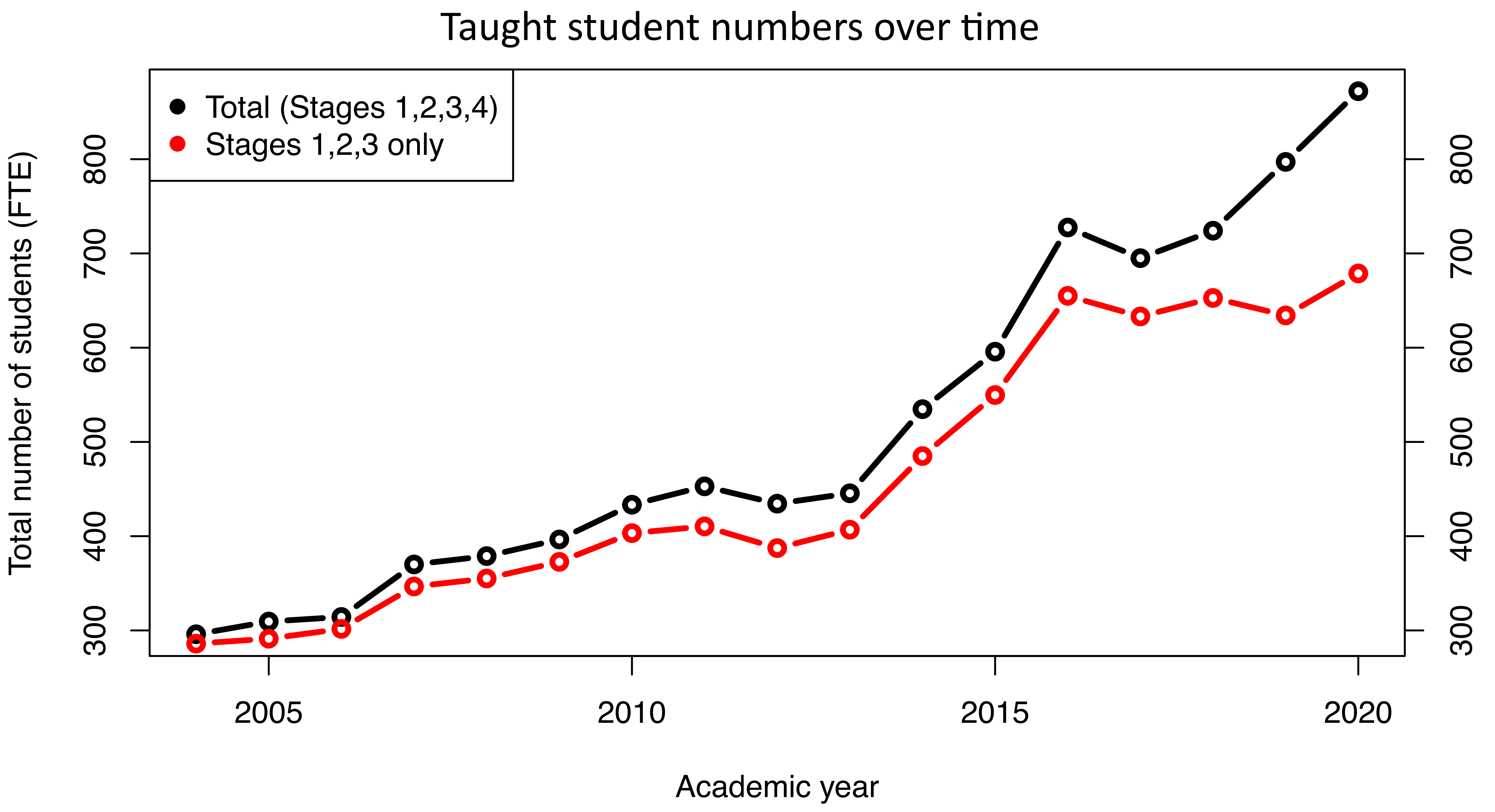
Connections across the University:

BSc and MSci in:

- Mathematics (also MMath)
- Mathematics and Data Science
- Mathematics with Finance
- Mathematics with Management
- Mathematics with Accounting
- Mathematics with Economics
- Mathematics and Physics

MSc in:

- Advanced mathematics
- Financial mathematics
- Mathematical modelling
- Statistics
- **Applied Data Science and Statistics (new!)**
- **ADSS + Modelling/Renewables etc (Penryn, new!)**
- **Weather and Climate Science (new!)**



Student numbers relentlessly increasing: tripled in last 15 years.
Up by 150 in last two years.

NSS: 5th in Russell group, 8th in Russell group in Guardian rankings.

Example of Teaching Innovation

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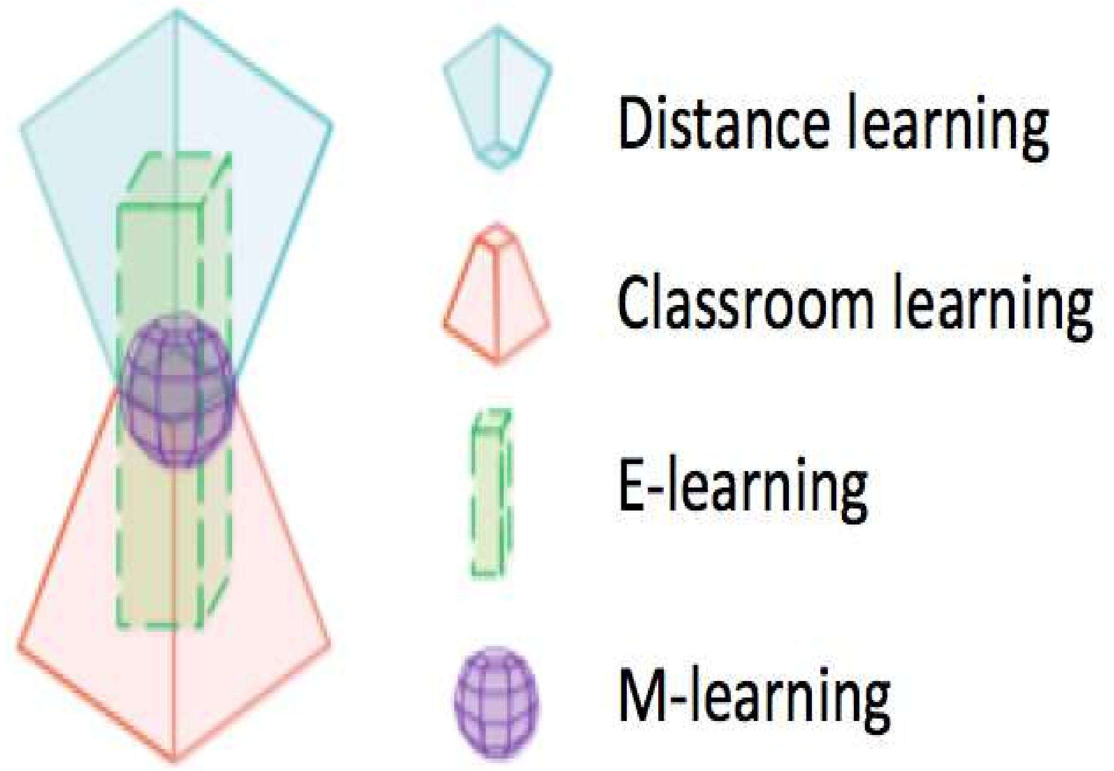
Competition: Imagining the Future of STEM Education

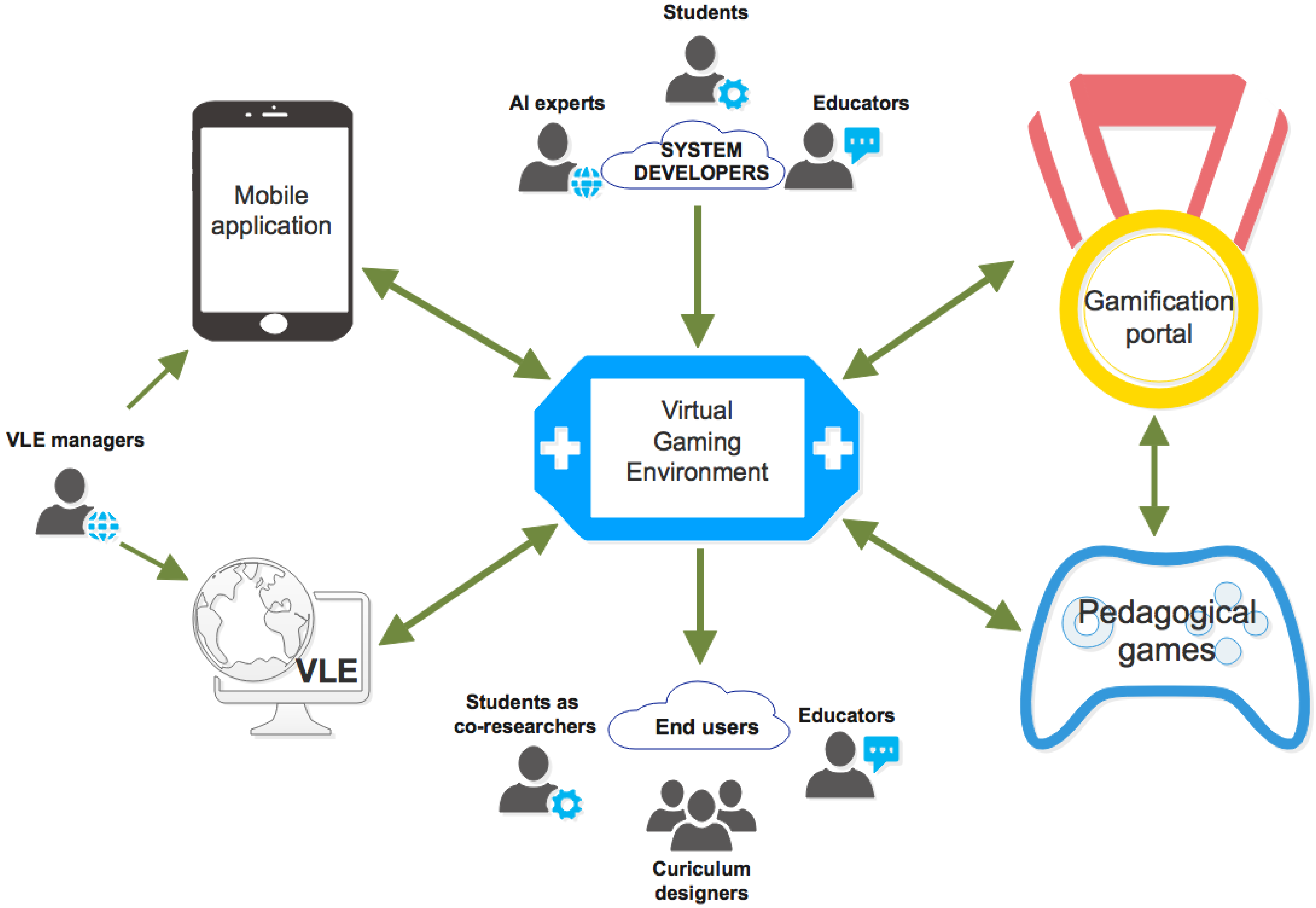
The National Academies of Sciences, Engineering, and Medicine.

1400 worldwide applications.
Honourable mention, University of Exeter, **Dr. Layal Hakim**

Collaboratively personalising STEM education through a virtual gaming environment.

- **Vision**: to combine traditional and new approaches to provided a new, integrated approach to STEM education.
- Distance learning, classroom learning, e-learning and mobile-learning.
- Embed course resources, computer aided assessments, and a gamification tool to enhance student's experience and collaboration.
- Brings together students from globally diverse educational backgrounds.





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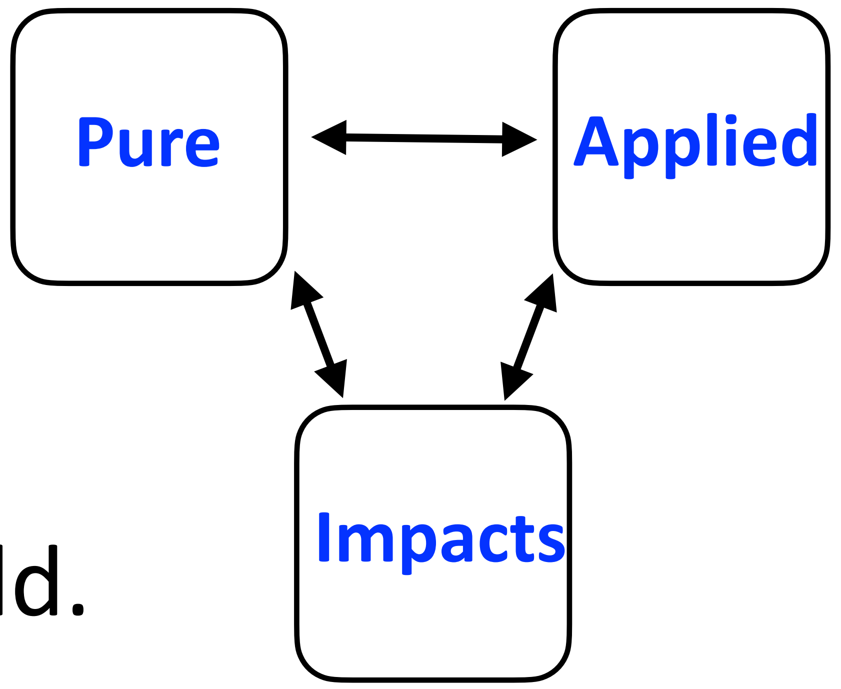
Challenges



- Better convince the world of our outstanding research and education.
  Increase our international presence and get the best students from anywhere in the world.

- Student numbers *increasing*, number of academic staff *falling* recently.
  How to maintain high quality teaching and research as SSR increases?

- Lack of office space – many shared offices, distribution across 3 buildings. (No labs.)
  Approaching physical capacity in teaching space.

- Teaching *and* Research, not Teaching *vs.* Research.
  Barriers in the reward system; E&R and E&S differences; drawbacks of SWARM.

- Large disciplinary range *within* the department: both a success and a challenge.
  Different metrics of success, not always recognised.

We are a thriving department with a trifecta of *pure* mathematics, *applied* mathematics, and *impacts* that reinforce and build on each other.
They are all strong, and they are all crucial to our success. We must build on that.